Untitled

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```r
# loading Grades data and preview
df = read.csv("Grades.csv")
head(df)
```

```
##   HW.Total Midterm1 Midterm2 Midterm3 Final Sex
## 1       87       78       42       50   123   M
## 2      139       65       84       50   110   M
## 3       56       71       87        0   139   F
## 4       74       85       66       57   137   M
## 5       82       86       75       70   131   F
## 6      137       70       77       67   159   M
```

Q.1 a) STAT 4 C03

```r
# modeling
model.a = lm("Final ~ Midterm1 + Midterm2 + Midterm3 + HW.Total", data=df)
# summarizing the model
summary(model.a)
```

```
##
## Call:
## lm(formula = "Final ~ Midterm1 + Midterm2 + Midterm3 + HW.Total",
##     data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -37.951  -5.939   1.363   8.637  24.726
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 41.79892   15.95681   2.620  0.01085 *
## Midterm1     0.48506    0.17358   2.794  0.00675 **
## Midterm2     0.60537    0.14230   4.254 6.57e-05 ***
## Midterm3     0.24732    0.10854   2.279  0.02584 *
## HW.Total     0.08206    0.04492   1.827  0.07213 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 12 on 68 degrees of freedom
## Multiple R-squared:  0.5459, Adjusted R-squared:  0.5192
## F-statistic: 20.44 on 4 and 68 DF,  p-value: 4.297e-11
```

First of all, based on the F-statistic's p-value, the model is clearly relevant and performs much better than a random model. This shows that the set of predictors explains a significant part of the variability in the final score.

Among all the covariates, Midterm2 is the most discriminating variable for predicting the Final score, with (*) **significance, followed by Midterm1, which is also strongly significant** (). This means that these two exams contribute the most to explaining the Final result.

Both the intercept and Midterm3 have (*) significance, which indicates that they have a weaker but still statistically meaningful effect on the Final score compared to Midterm1 and Midterm2.

Finally, HW.Total is not very significant, meaning that it does not really help to explain the Final score once the midterm results are already included in the model.

Overall, the model shows that performance on the midterms, especially Midterm2 and Midterm1, is the main driver of the Final score, while homework has only a limited additional contribution.

Q.1 b)

```
# creating the new data point
new = data.frame(
  HW.Total = 85,
  Midterm1 = 70,
  Midterm2 = 70,
  Midterm3 = 70
)

# predicting the mean final score with 95% confidence
predict(model.a, newdata=new, interval="confidence", level=0.95)
```

```
##        fit      lwr      upr
## 1 142.4162 134.0899 150.7425
```

Q.1 c)

```
# adding the midterm mean column in the dataframe
df$avg.midterm = (df$Midterm1 + df$Midterm2 + df$Midterm3)/3
# modeling
model.c = lm(Final ~ HW.Total + avg.midterm, data = df)
# summarizing the model
summary(model.c)
```

```
##
## Call:
## lm(formula = Final ~ HW.Total + avg.midterm, data = df)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.594  -6.964   1.190   8.103  21.797
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 55.80559   13.31790   4.190  8.0e-05 ***
## HW.Total     0.07980    0.04517   1.766   0.0817 .
## avg.midterm  1.20145    0.18031   6.663  5.1e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.13 on 70 degrees of freedom
## Multiple R-squared:  0.523,  Adjusted R-squared:  0.5093
## F-statistic: 38.37 on 2 and 70 DF,  p-value: 5.608e-12
```

```
# Wald test
# Taking the average means to define R in order to have beta1 = beta2 = beta 3
R = rbind(
  c(0, 1,  -1, 0,  0),
  c(0, 0,  1,  -1, 0)
)
# reference: Jan 21 R code
q=nrow(R)
Rbhat=R%*%coef(model.a)
Vbhat=summary(model.a)$cov.unscaled
VRbhat=R%*%Vbhat%*%t(R)
MSE=summary(model.a)$sigma^2
Fstat1=(t(Rbhat)%*%solve(VRbhat)%*%Rbhat)/(q*MSE)

c("F"=Fstat1, "p.val"=pf(Fstat1,q,model.a$df.residual, lower=F))
```

```
##         F     p.val
## 1.7196504 0.1868265
```

```
# Likelihood Ratio Test
anova(model.a, model.c)
```

```
## Analysis of Variance Table
##
## Model 1: Final ~ Midterm1 + Midterm2 + Midterm3 + HW.Total
## Model 2: Final ~ HW.Total + avg.midterm
##   Res.Df     RSS Df Sum of Sq      F Pr(>F)
## 1     68  9798.1
## 2     70 10293.6 -2   -495.57 1.7197 0.1868
```

The Wald and LRT tests give the same result. With F = 1.7197 and p-value = 0.1868

Q.1 d)

```r
# new data point
new = data.frame(
  HW.Total = 85,
  avg.midterm = 70
)
# prediction
predict(model.c, newdata=new, interval="confidence", level=0.95)
```

```
##        fit      lwr      upr
## 1 146.6898 139.9228 153.4568
```

The reduced model's interval is slightly shifted upward compared to the full mode.